

Bisection vs Anderson–Björck:

A Comparison of Root-finding Methods

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These slides cover 4 root-finding methods:

1. Bisection
2. Regula falsi “method of false position”
3. Illinois method (Snyder, 1953; Dowell & Jarratt, 1971)
4. Anderson–Björck (Anderson & Björck, 1973)

I will assume you know the concept of bisection, so that will be a quick review/reference point for us

Why even do this?-

Well, why even do anything? If you're going to do something, do it well.

More importantly, when it's 3:30am and you're trying to speed through a calibration, each iteration takes years off your life and decades off your rapidly dwindling sanity

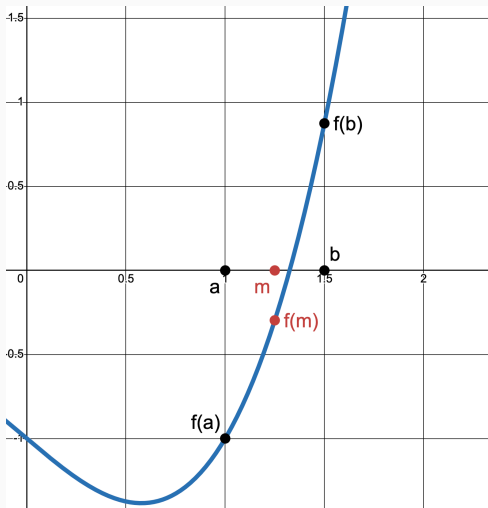
- You'd like to shrink the initial brackets, but when targeting strong asymmetries (e.g. size distributions), things don't behave as nicely as you'd like
- Widening the brackets adds 1, maybe 2 iters, but you are no longer a sane person that properly allocates your time

Suppose I wanted to find the root of $f(x)$, which is continuous between a and b :

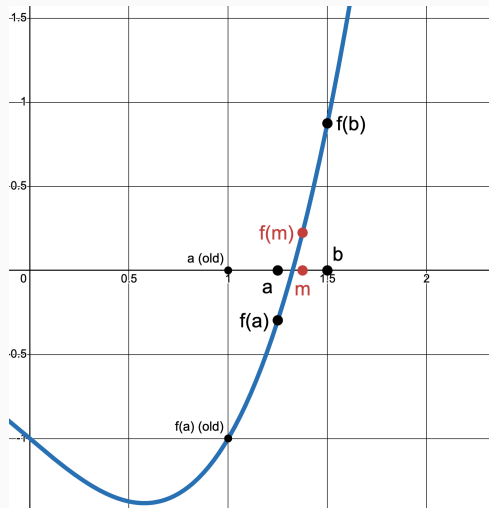
1. Pick initial brackets (a and b) where $f(a)$ and $f(b)$ have opposite signs
2. Find the midpoint of a and b , $m = \frac{a+b}{2}$ and take $f(m)$
3. If $f(m)$ has the same sign as $f(a)$, replace a with m
 - Or, replace b with m if $f(m)$ has the same sign as $f(b)$
4. Continue until $|a - b| < \text{desired_tol}$. Then m sufficiently approximates the root

Next slide: example with $f(x) = x^3 - x - 1$ and initial $a = 1$ and $b = 1.5$

Bisection: Steps 1-3 (All graphs made with Desmos)



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Simple Root-finding

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Bisection: Steps 4 - N

Now $m = 1.375$ becomes the new b and the process repeats until convergence

- Determined by $|a - b| < \epsilon$, where ϵ is sufficiently small and chosen by the modeler

This is a perfectly valid method, but we can make it faster with the “false position”

We use bisection, or various other numerical methods, because economics is hard

- What is simple is finding the root of a line

We can use the root of a line to better update our new a or b point:

1. Instead of the midpoint of a and b , draw a line between $f(a)$ and $f(b)$
2. Call the root of this line c and take $f(c)$
 - I'm using c since this is no longer a midpoint, but call it m if you like
3. If $f(c)$ has the same sign as $f(a)$, replace a with c
 - Or, replace b with c if $f(c)$ has the same sign as $f(b)$
4. Continue until $|a - b| < \text{desired_tol}$. Then c approximates the root (more latter?)

Building the Line

To draw our line, linearly interpolate (and rearrange a bit) $f(a)$ and $f(b)$:

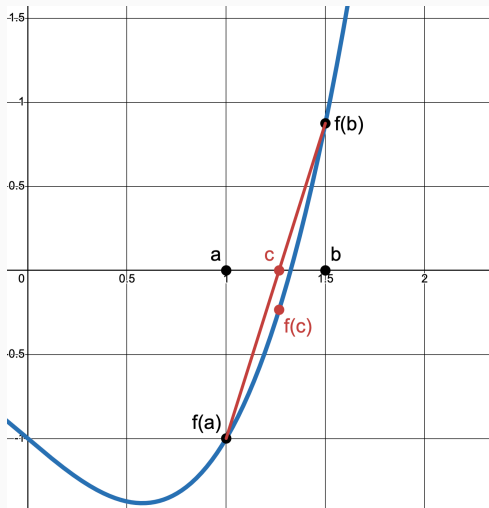
$$c = b - \frac{f(b) \times (b - a)}{f(b) - f(a)}$$

Optional: set a condition to use the smaller $|f(val)|$. If $|f(a)| < |f(b)|$, use:

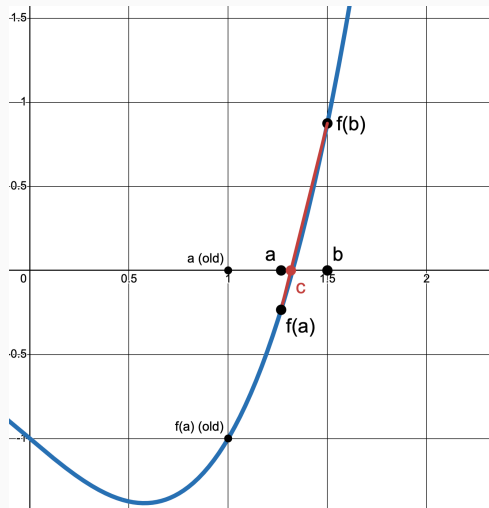
$$c = a - \frac{f(a) \times (b - a)}{f(b) - f(a)}$$

- This likely won't matter outside of floating-point dust

Regula Falsi: Steps 1-4



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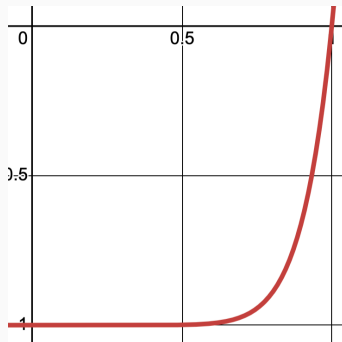
Simple Root-finding

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Issues?

Consider the case of $f(x) = x^{10} - 1$ on $[0, 1.3]$

- $f(x)$ is flat near 0 and blows up near 1.3
- Our line always crosses the axis around the left end
- $b = 1.3$ is retained forever and a crawls towards 1 (true root)
- *Why can't $|a - b|$ work here? Hint: think about 0.3*



The Illinois Method

The Illinois method aims to correct this issue

- First as technical note by Snyder (1953), then formalized by Dowell & Jarratt (1971)

WLOG, if $\text{sign } f(c) = \text{sign } f(a)$, then the root is in $[c, b]$

- c is the new a ; b remains the same
- No difference from regula falsi yet

The change comes from how we “dislodge” $f(b)$...

We need $f(b)$ to move closer to 0 so we just halve it

Take the same regula falsi algorithm we already built, modified with:

- If $\text{sign } f(c) = \text{sign } f(a)$ twice in a row, $f(b) = \frac{f(b)}{2}$
- If it happens again on the next iter, halve again...
- *Everything else remains the same as in regula falsi*
- Convergence guaranteed (halving does not change sign; stale bracket replaced)
 - ▷ Order of convergence ≈ 1.442 (Dowell & Jarratt, 1971)

Note: some always halve the non-moving bound on every iteration. I haven't seen a noticeable difference when I do this.

Regula Falsi Failure and Illinois Correction for $f(x) = x^{10} - 1$

Regula Falsi:

```
iter=1 a=0.0 b=1.3 c=0.0942995953723274
iter=2 a=0.0942995953723274 b=1.3 c=0.18175887251907943
iter=3 a=0.18175887251907943 b=1.3 c=0.26287401252030435
iter=4 a=0.26287401252030435 b=1.3 c=0.3381051033222695
iter=5 a=0.3381051033222695 b=1.3 c=0.4078779165927524
iter=6 a=0.4078779165927524 b=1.3 c=0.47258315356239
iter=7 a=0.47258315356239 b=1.3 c=0.5325715106320142
iter=8 a=0.5325715106320142 b=1.3 c=0.5881445691706799
iter=9 a=0.5881445691706799 b=1.3 c=0.6395439707682159
iter=10 a=0.6395439707682159 b=1.3 c=0.6869431667412391
iter=11 a=0.6869431667412391 b=1.3 c=0.7304464366929381
iter=12 a=0.7304464366929381 b=1.3 c=0.7700987444968075
iter=13 a=0.7700987444968075 b=1.3 c=0.8059075090169241
iter=14 a=0.8059075090169241 b=1.3 c=0.8378738914244527
iter=15 a=0.8378738914244527 b=1.3 c=0.8660276320873561
iter=16 a=0.8660276320873561 b=1.3 c=0.8904572813509999
iter=17 a=0.8904572813509999 b=1.3 c=0.911328251308228
iter=18 a=0.911328251308228 b=1.3 c=0.9288846695992143
iter=19 a=0.9288846695992143 b=1.3 c=0.9434360517492835
iter=20 a=0.9434360517492835 b=1.3 c=0.9553339751921469
iter=21 a=0.9553339751921469 b=1.3 c=0.9649455723416893
iter=22 a=0.9649455723416893 b=1.3 c=0.9726296888054463
iter=23 a=0.9726296888054463 b=1.3 c=0.9787191336334624
iter=24 a=0.9787191336334624 b=1.3 c=0.9825600573333043
```

*Max iter hit at 100 using $|f(c)| < \epsilon = 1e-12$

Illinois:

```
iter=1 a=0.0 b=1.3 c=0.0942995953723274
iter=2 a=0.0942995953723274 b=1.3 c=0.18175887251907943
iter=3 a=0.18175887251907943 b=1.3 c=0.3330171567671213
iter=4 a=0.3330171567671213 b=1.3 c=0.5634423147022629
iter=5 a=0.5634423147022629 b=1.3 c=0.8463635731395354
iter=6 a=0.8463635731395354 b=1.3 c=1.074910177068493
iter=7 a=0.8463635731395354 b=1.074910177068493 c=0.9454923183277778
iter=8 a=0.9454923183277778 b=1.074910177068493 c=0.9828011093189349
iter=9 a=0.9828011093189349 b=1.074910177068493 c=1.0040954923602121
iter=10 a=0.9828011093189349 b=1.0040954923602121 c=0.9996755374546944
iter=11 a=0.9996755374546944 b=1.0040954923602121 c=0.9999940615677148
iter=12 a=0.9999940615677148 b=1.0040954923602121 c=1.000005704633256
iter=13 a=0.9999940615677148 b=1.000005704633256 c=0.9999999998475554
iter=14 a=0.9999999998475554 b=1.000005704633256 c=0.999999999999961
```

The Anderson–Björck Adjustment

The Illinois method is great, but it can be sped up with a dynamic adjustment factor

WLOG, if $\text{sign } f(c) = \text{sign } f(a)$, Anderson and Björck use $k \times f(b)$, where:

$$k' = 1 - \frac{f(c)}{f(a)},$$
$$k = \begin{cases} k' & \text{if } k' > 0, \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

Updating here happens each time, not after two matching signs in a row

- *Everything else still remains the same as in regula falsi*
 - ▷ Order of convergence ≈ 1.7 (Anderson & Björck, 1973)

From before:

$$k' = 1 - \frac{f(c)}{f(a)},$$
$$k = \begin{cases} k' & \text{if } k' > 0, \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

If $k' < 0$, we want to avoid using it and “artificially” changing a sign with this weight

- *But what does this tell us about the step we just took?*
- Fallback is simply Illinois

My JMP D-StSt: Bisection vs Anderson-Björck

Bisection:

METHOD 1: BISECTION bracket [5, 10], xtol = 1e-07 on the price					
iter	pval	wage	C	g=C-1/p	
1	5.0000000	0.428000	0.030432	-1.70e-01	
2	10.0000000	0.214000	0.568592	+4.69e-01	
3	7.5000000	0.285333	0.167265	+3.39e-02	
4	6.2500000	0.342400	0.077432	-8.26e-02	
5	6.8750000	0.311273	0.115721	-2.97e-02	
6	7.1875000	0.297739	0.139706	+5.76e-04	
7	7.0312500	0.304356	0.127275	-1.49e-02	
8	7.1093750	0.301011	0.133378	-7.28e-03	
9	7.1484375	0.299366	0.136514	-3.38e-03	
10	7.1679688	0.298550	0.138103	-1.41e-03	
11	7.1777344	0.298144	0.138903	-4.17e-04	
12	7.1826172	0.297942	0.139304	+7.92e-05	
13	7.1801758	0.298043	0.139104	-1.69e-04	
14	7.1813965	0.297992	0.139204	-4.48e-05	
15	7.1820068	0.297967	0.139254	+1.72e-05	
16	7.1817017	0.297980	0.139229	-1.38e-05	
17	7.1818542	0.297973	0.139241	+1.69e-06	
18	7.1817780	0.297976	0.139235	-6.07e-06	
19	7.1818161	0.297975	0.139238	-2.19e-06	
20	7.1818352	0.297974	0.139240	-2.52e-07	
21	7.1818447	0.297974	0.139241	+7.17e-07	
22	7.1818399	0.297974	0.139240	+2.32e-07	
23	7.1818376	0.297974	0.139240	-1.02e-08	
24	7.1818388	0.297974	0.139240	+1.11e-07	
25	7.1818382	0.297974	0.139240	+5.04e-08	
26	7.1818379	0.297974	0.139240	+2.01e-08	
27	7.1818377	0.297974	0.139240	+4.95e-09	
28	7.1818376	0.297974	0.139240	-2.62e-09	

METHOD 1: BISECTION: p = 7.1818376 28 solves total 161.22 seconds

Anderson-Björck:

METHOD 3: ANDERSON-BJÖRCK bracket [5, 10], xtol = 1e-07 on the price					
iter	pval	wage	C	g=C-1/p	
1	5.0000000	0.428000	0.030432	-1.70e-01	
2	10.0000000	0.214000	0.568592	+4.69e-01	
3	6.3285720	0.338149	0.081606	-7.64e-02	
4	7.1688375	0.298514	0.138174	-1.32e-03	
5	7.1835141	0.297904	0.139378	+1.70e-04	
6	7.1818333	0.297974	0.139240	-4.41e-07	
7	7.1818377	0.297974	0.139240	+1.85e-11	
8	7.1818377	0.297974	0.139240	-1.67e-16	

METHOD 3: ANDERSON-BJÖRCK: p = 7.1818377 8 solves total 46.01 seconds

*Illinois took roughly 1 minute.

Other Methods?

Many other weights exist, along with a multitude of completely different methods, which may be faster

HOWEVER, Anderson–Björck has three benefits in one: speed, [relative] ease of learning, and guaranteed to work

On Brent (1971), I think this is what 'fzero' does in MATLAB, but YOU'RE not using pre-made computations anyway, right?? Regardless, the decision tree you'll have to follow for Brent is annoying and the path can move a lot with different warm starts. Not something your K-S regression coefficients will like without a strong tolerance. You do not want to debug this at 4:00am.

References

-  Anderson, N. & Å. Björck (1973). “A New High Order Method of Regula Falsi Type for Computing a Root of an Equation”. In: *BIT* 13.3, pp. 253–264. DOI: [10.1007/BF01951936](https://doi.org/10.1007/BF01951936).
-  Brent, R. P. (1971). “An Algorithm with Guaranteed Convergence for Finding a Zero of a Function”. In: *The Computer Journal* 14.4, pp. 422–425. DOI: [10.1093/comjnl/14.4.422](https://doi.org/10.1093/comjnl/14.4.422).
-  Dowell, Mark & Peter Jarratt (1971). “A Modified Regula Falsi Method for Computing the Root of an Equation”. In: *BIT* 11.2, pp. 168–174. DOI: [10.1007/BF01934364](https://doi.org/10.1007/BF01934364).
-  Snyder, James N. (1953). *Inverse interpolation, a real root of $f(x) = 0$* . ILLIAC I Library Routine H1-71. University of Illinois Digital Computer Laboratory, p. 4.

Graphs: <https://www.desmos.com/calculator>